

Renewable Energy Logistics & Supply Chain Performance Analysis

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A data-driven analysis of renewable-energy logistics operations, examining supplier performance, route delays, freight-cost variance, material damage, transportation distance, and corrective-action priorities. The document presents the key business questions, SQL analysis results, supporting tables, and actionable insights to identify operational inefficiencies and improve delivery reliability, cost control, and supply-chain performance.

Business Questions

1. Which suppliers are performing best across on-time delivery, damage rate, and freight cost?

	supplier text	on_time_rate numeric	avg_damage_rate numeric	total_freight_cost numeric
1	SunCore Energy	50.769	0.021	62499256
2	SolarTech Supplies	48.438	0.019	64914120
3	EcoPower Logistics	44.828	0.018	57807682
4	GreenVolt Nigeria	44.643	0.018	54214496
5	PowerGrid Solutions	35.088	0.018	56168662

Insight: *Delivery reliability varies considerably by supplier, with on-time rates ranging from 35.1% to 50.8%. PowerGrid Solutions records the lowest on-time rate, while SolarTech Supplies incurs the highest total freight cost at ₦64.9M. Damage rates are comparatively consistent across suppliers, ranging from 1.8% to 2.1%.*

2. Which origin–destination routes experience the highest delivery delays and cost overruns?

	route text	shipments bigint	avg_delay_days numeric	avg_cost_ouerrun numeric	total_cost_ouerrun numeric
1	Edo→Kwara	2	7.00	18555.00	37110
2	Rivers→Niger	1	7.00	14903.00	14903
3	Oyo→Anambra	1	7.00	-112461.00	-112461
4	Lagos→Delta	2	6.00	5147.00	10294
5	Edo→Ogun	2	5.50	-11451.00	-22902
6	Oyo→Cross River	3	5.33	-52965.67	-158897
7	Kaduna→Cross River	1	5.00	-20865.00	-20865
8	Plateau→Kano	2	5.00	-27306.00	-54612
9	Niger→Kaduna	1	5.00	-108316.00	-108316
10	Niger→Ogun	1	4.00	43655.00	43655

Insight: Route performance shows that delivery delays and freight cost overruns do not always move together. Edo → Kwara and Rivers → Niger recorded the highest average delays at 7 days while also exceeding budget, making them immediate routes for operational review. However, Niger → Ogun recorded the largest average cost overrun of ₦43,655 despite a lower 4-day delay, suggesting that freight pricing and route costs require separate investigation from transit performance.

3. What factors are most associated with late deliveries, particularly distance, priority, supplier, and material?

	material_category text	shipments bigint	late_delivery_rate numeric	avg_delay_days numeric
1	Mounting Structures	41	68.29	1.76
2	Solar Water Pump	32	59.38	1.84
3	Cables & Protection	41	58.54	1.56
4	Solar Panels	43	55.81	1.33
5	Monitoring Equipment	33	54.55	1.91
6	Lithium Batteries	41	48.78	1.54
7	Hybrid Inverters	42	47.62	1.48
8	Transformers	27	44.44	1.33

Insight: Material category is associated with meaningful differences in delivery reliability. Mounting Structures record the highest late-delivery rate at 68.3%, followed by Solar Water Pumps at 59.4%, while Transformers have the lowest at 44.4%. Monitoring Equipment records the highest average delay at 1.91 days, suggesting that frequency of late deliveries and the severity of delays may have different underlying drivers.

4. Which renewable-energy materials have the highest damage rates and shipment losses?

	material_category text	shipments bigint	units_shipped bigint	units_damaged bigint	damage_rate_pct numeric	shipment_losses bigint
1	Transformers	27	2474	52	2.10	52
2	Solar Water Pump	32	2598	54	2.08	54
3	Monitoring Equipment	33	2688	55	2.05	55
4	Cables & Protection	41	3748	75	2.00	75
5	Hybrid Inverters	42	4077	80	1.96	80
6	Lithium Batteries	41	3993	78	1.95	78
7	Solar Panels	43	3616	58	1.60	58
8	Mounting Structures	41	3032	48	1.58	48

Insight: Material damage rates are relatively consistent, ranging from 1.58% to 2.10%. Transformers record the highest damage rate at 2.10%, followed by Solar Water Pumps at 2.08%. However, Hybrid Inverters account for the highest absolute number of damaged units at 80, reflecting their larger shipment volume. This suggests that

corrective action should consider both damage rates and total material losses when prioritizing materials for review.

5. Which suppliers and routes consistently exceed their budgeted freight costs?

	supplier text	shipments bigint	over_budget_shipments bigint	over_budget_rate_pct numeric	avg_cost_variance_ngn numeric	total_cost_variance_ngn numeric
1	SunCore Energy	65	33	50.77	-4023.86	-261551
2	GreenVolt Nigeria	56	23	41.07	-18708.46	-1047674
3	PowerGrid Solutions	57	22	38.60	-24937.88	-1421459
4	SolarTech Supplies	64	24	37.50	-26437.22	-1691982
5	EcoPower Logistics	58	16	27.59	-39296.91	-2279221

Insight: Supplier-level cost performance shows that individual shipment overruns are common, but all suppliers remain below their aggregate freight budgets. SunCore Energy records the highest over-budget shipment rate at 50.77%, while EcoPower Logistics has the lowest at 27.59% and the largest overall budget saving of ₦2.28M. This suggests management should investigate the frequency and drivers of individual shipment overruns rather than treating supplier-level spending as uniformly over budget.

6. Does longer transportation distance lead to higher freight costs, longer transit times, or greater material damage?

	distance_km integer	freight_cost_ngn bigint
1	25	1066770
2	26	1699264
3	28	150612
4	31	391618
5	32	841536
6	35	1637052
7	36	306135
8	37	1122649
9	38	1324060
10	42	130698

Insight: Transportation distance appears to be positively associated with freight costs, but the relationship is not uniform. Shipments covering similar distances can have

substantially different freight costs, indicating that factors beyond distance may influence logistics expenditure. Further correlation analysis is required to determine whether distance is also associated with longer transit times or higher material damage.

7. Which suppliers, routes, and materials should management prioritize for corrective action to improve delivery reliability and reduce logistics costs?

A. Suppliers to Prioritize

	supplier text	shipments bigint	late_delivery_rate_pct numeric	avg_delay_days numeric	over_budget_rate_pct numeric	total_cost_variance_ngn numeric
1	PowerGrid Solutions	57	64.91	1.88	38.60	-1421459
2	GreenVolt Nigeria	56	55.36	1.25	41.07	-1047674
3	EcoPower Logistics	58	55.17	1.47	27.59	-2279221
4	SolarTech Supplies	64	51.56	1.48	37.50	-1691982
5	SunCore Energy	65	49.23	1.83	50.77	-261551

Insight: *Supplier performance reveals different operational issues rather than a single cost-and-reliability problem. PowerGrid Solutions records the highest late-delivery rate at 64.9%, while SunCore Energy has the highest proportion of shipments exceeding budget at 50.8%. GreenVolt Nigeria also has a relatively high over-budget rate of 41.1%. Although all suppliers remain below their aggregate freight budgets, the frequency of late deliveries and individual shipment overruns suggests that corrective action should focus on improving delivery reliability and controlling shipment-level cost deviations.*

B. Routes to Prioritize

	route text	shipments bigint	avg_delay_days numeric	late_delivery_rate_pct numeric	avg_cost_variance_ngn numeric	total_cost_variance_ngn numeric
1	Kwara → Abia	5	2.40	60.00	-6585.20	-32926
2	Abia → Oyo	5	3.40	60.00	-51043.60	-255218
3	Abia → Cross River	6	0.83	33.33	-11901.83	-71411

Insight: Route analysis identifies Kwara → Abia and Abia → Oyo as the main delivery-reliability concerns, with both recording 60% late deliveries. Abia → Oyo has the highest average delay at 3.4 days, compared with 2.4 days for Kwara → Abia. Although these routes remain below budget overall, their delivery performance warrants investigation into carrier scheduling, route conditions, and transit handoffs.

C. Materials to Prioritize

	material_category text	shipments bigint	avg_delay_days numeric	late_delivery_rate_pct numeric	damage_rate_pct numeric	units_damaged bigint	total_cost_variance_ngn numeric
1	Mounting Structures	41	1.76	68.29	1.58	48	-1507249
2	Solar Water Pump	32	1.84	59.38	2.08	54	-969116
3	Cables & Protection	41	1.56	58.54	2.00	75	-564643
4	Solar Panels	43	1.33	55.81	1.60	58	-875993
5	Monitoring Equipment	33	1.91	54.55	2.05	55	-584229
6	Lithium Batteries	41	1.54	48.78	1.95	78	-1058353
7	Hybrid Inverters	42	1.48	47.62	1.96	80	-1149971
8	Transformers	27	1.33	44.44	2.10	52	7667

Insight: Mounting Structures should receive the strongest delivery-focused intervention, while Cables & Protection, Monitoring Equipment, Solar Water Pumps and Hybrid Inverters warrant attention because of the combined delivery and damage exposure.

One important distinction: Hybrid Inverters have the highest number of damaged units (80), while Transformers have the highest damage rate (2.10%). Management should therefore track both damage rate and absolute damaged units rather than relying on either metric alone.

Also, all materials except Transformers have a negative total cost variance, meaning they are under aggregate freight budget despite some individual shipment overruns.

D. Corrective Action

	supplier text	shipments bigint	late_delivery_rate_pct numeric	avg_delay_days numeric	over_budget_rate_pct numeric	total_cost_variance_ngn numeric	corrective_action text
1	SunCore Energy	65	49.23	1.83	50.77	-261551	High Priority
2	PowerGrid Solutions	57	64.91	1.88	38.60	-1421459	Medium Priority
3	GreenVolt Nigeria	56	55.36	1.25	41.07	-1047674	Medium Priority
4	EcoPower Logistics	58	55.17	1.47	27.59	-2279221	Medium Priority
5	SolarTech Supplies	64	51.56	1.48	37.50	-1691982	Medium Priority

Insight: *SunCore Energy is classified as High Priority under the defined corrective-action thresholds because of its combination of a 49.23% late-delivery rate and 50.77% over-budget shipment rate. PowerGrid Solutions records the highest late-delivery rate at 64.91%, making delivery reliability a key area for review.*